Hypatia's work was later expanded on by eminent scientists, including Isaac Newton, who used conic sections to realize his theories of planetary movement (see pp.76–81).

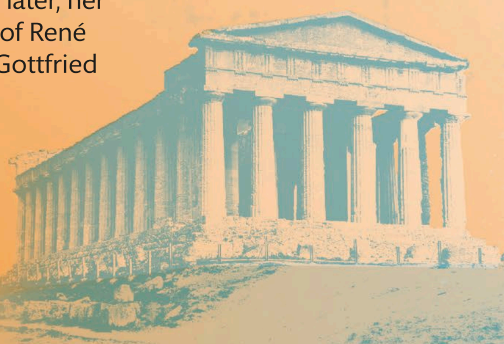
Skilled in algebra, Hypatia wrote a commentary on the *Arithmetica of Diophantus*, helping to explain the principles of solving algebraic equations. Parts of her commentary are thought to have been reworked into a later Arab edition of *Diophantus*, which survived the early Middle Ages and reemerged in the Renaissance, influencing Pierre Fermat's development of modern number theory.

In addition to her mathematics work, Hypatia has been associated with the development of several mechanical devices. In the preserved letters of Synesius of Cyrene—later Bishop of Ptolemais, one of her most devoted students—are instructions for constructing an astrolabe, a navigational instrument whose design was advanced enough to measure the positions of the planets, the stars, and the Sun. Synesius claimed that Hypatia invented the astrolabe, although it is more likely that she made improvements to the existing design (see box). Another device attributed to Hypatia in the letters is a hydrometer for measuring the specific gravity of a liquid.

Enduring legacy

Regarded as the most celebrated female scientist of the ancient world, Hypatia's advances were significant not only because of her gender, but also because she was one of the last scientists of note before the early Middle Ages, when scientific endeavors in Europe were largely stamped out by Christian zealots who believed math and science to be heresy. Hypatia lived during the final years of the Roman Empire, and her violent death is regarded as a result of growing Christian fervor against rational thought and those, like Hypatia, who were pagans. While scholarship might have gone into decline after Hypatia's death, centuries later, her ideas were to inform the work of René Descartes, Isaac Newton, and Gottfried Leibniz, among others.

> "Hypatia ... made such attainments in literature and science, as to **far surpass all the philosophers of her own time.**"
>
> **Socrates**, c.450 CE



An exceptional educator, *Hypatia was adored by her students. She lectured widely and popularized interest in mathematics and astronomy.*